

Academic Positions

- 2025– Assistant Professor (Tenure Track), Hebrew University Business School, Department of Operations Research.
- 2024– Research Affiliate, Massachusetts Institute of Technology, Intelligent Transportation Systems (ITS) Laboratory, Department of Civil and Environmental Engineering.
- 2024–2025 Postdoctoral Researcher, University of Toronto, Rotman School of Management, Faculty of Operations Management and Statistics.
- 2023–2024 Postdoctoral Researcher, Massachusetts Institute of Technology, Intelligent Transportation Systems (ITS) Laboratory, Department of Civil and Environmental Engineering.

Education

PhD in Operations Research, The Hebrew University of Jerusalem, *with honors (Hans Wiener Prize for outstanding doctoral degree)*.

Advisor: Professor Nicole Adler. Evaluating transportation markets through game theory.

Master Thesis in Industrial Engineering, Ben Gurion University.

Advisor: Professor Arie Gavious. Investing in research and teaching facilities in an academic institute - an optimal control and differential game model.

M.B.A. in Finance and Accounting, The Hebrew University of Jerusalem, *magna cum laude*.

B.Sc. in Industrial Engineering and Management, The Jerusalem College of Engineering, *summa cum laude*.

Grants

- 2023 MIT Portugal Program (MPP) Seed Grant, \$100,000.
- 2021 The Israeli Smart Transportation Research Center (ISTRC), \$30,000.
- 2018–2022 The Hebrew University of Jerusalem, University President's Full Scholarship of Distinction, \$100,000.

Awards

- 2023 Hans Wiener Prize for outstanding doctoral degree.
- 2018–2022 The Hebrew University of Jerusalem, University President’s Full Scholarship of Distinction.
- 2021 The Gal-Edd foundation scientific publication award.
- 2021 Best scientific presentation, The Israeli Smart Transportation Research Center (ISTRC) 2nd student conference.
- 2020–2021 Award for Excellence in Teaching at Azrieli College of Engineering.
- 2009 Dean’s List Award at The Hebrew University of Jerusalem. Graduated *magna cum laude*, with a GPA of 94.96.
- 2003–2006 Dean’s List Award in all years and Academic Excellence Scholarship in all years at The Jerusalem College of Engineering. Graduated *summa cum laude* 1st in class year with a GPA of 95.15

Publications

Brudner, A., Gaviols, A. (2024) A Dynamic model of investment in research and teaching facilities in academic institutions. *Annals of Operations Research*.

Brudner, A. (2023). On the management of residential on-street parking: Policies and repercussions. *Transport Policy*, 138, 94-107.

Adler, N., Brudner, A., Gallotti, R., Privitera, F., & Ramasco, J. J. (2022). Does big data help answer big questions? The case of airport catchment areas & competition. *Transportation Research Part B: Methodological*, 166, 444-467.

Adler, N., Brudner, A., & Proost, S. (2021). A review of transport market modeling using game-theoretic principles. *European Journal of Operational Research*, 291(3), 808-829.

Working Papers

Brudner, A., Patricio, A., Santos, G., Antunes, A. & Ben-Akiva, M. How driving automation will save demand responsive transit. (**Submitted for publication**)

Brudner, A., & Adler, N. Regulating automated vehicle fleets under market power.

Blum, E., Brudner, A., García-Albertos, p., & Adler, N. Impact of autonomous vehicles on public transport in medium-haul markets

Teaching Experience

- 2024 Kaufman Teaching Certificate Program (KTCP), Massachusetts Institute of Technology (MIT)
Comprehensive training in pedagogy, course design and student-centered teaching methodologies.
- 2016– The Hebrew University of Jerusalem, Hebrew University Business School, Lecturer
Courses: Operations Management, Quantitative Models (in English)
- 2008–2022 The Jerusalem College of Engineering, Department of Industrial Engineering, Teaching Fellow and Course Instructor
Courses: Information Systems Analysis and Planning, Statistics, Operations Research, Deterministic and Stochastic Models, Engineering Optimization, Database Management and SQL, Control and Quality Assurance, Intro to Industrial Engineering and Management.
Award for Excellence in Teaching (2020,2021)
- 2011–2017 The Jerusalem College of Engineering, Department of Industrial Engineering, Lecturer
Courses: Methods Engineering, Control and Quality Assurance
- 2011–2017 The Jerusalem College of Engineering, Department of Industrial Engineering, Academic Supervisor for Final Projects
Supervises projects in Operations Research and Information Systems

Academic Services

Peer Reviewer: i) European Journal of Operational Research. ii) Transportation Research Part A: Policy and Practice. iii) Transportation Research Part B: Methodological. iv) Journal of Air Transport Management. v) Journal of Urban Affairs. vi) Systems & Control Letters.

Event Organization: i) Organizing the second ISTRC student conference, 2021. ii) Leading the Intel Big Data Initiative, 2018.

Conferences, Presentation, Seminars and Workshops

- 2024 "Brudner, A., Patricio, A., Santos, G., Antunes, A. & Ben-Akiva, M. How Driving Automation Will Save Demand Responsive Transit."
Institute for Operations Research and the Management Sciences (INFORMS) Annual Meeting in Seattle, WA, USA (October 2024) and at the Transportation Research Board (TRB) (**Forthcoming** January 2025)
- 2022 "Adler, Nicole and Brudner, Amir: Creating equitable transport markets with AVs"
Annual Conference of the International Transportation Economics Association (ITEA) (June 2022)

- 2021 "Adler, Nicole and Brudner, Amir: Creating equitable transport markets with AVs"
Presented at the 2021 Institute for Operations Research and the Management Sciences (INFORMS) Annual Meeting in Anaheim, CA, USA (October 2021) and at the Operations and Research Society of Israel (ORSIS) annual meeting, Technion, Haifa, Israel (August 2021)
- 2019 "Brudner, Amir; Gallotti, Riccardo; Privitera, Filippo; J. Ramasco, Jose and Adler, Nicole: Big data and platform competition: The case of a multi-airport system"
Presented at the Annual School and Conference of the International Transportation Economics Association (ITEA), Paris School of Economics and at the 30th European Conference on Operational Research (EURO), Dublin, Ireland (June 2019)
- 2018 "Adler, Nicole and Brudner, Amir: Estimating High-Speed Rail and Airport Catchment Areas"
Presented at the 14th International Conference on Industrial Logistics (ICIL), Beer-Sheva, Israel (May 2018)
- 2017–2019 Air Transport and Regional Development (ATARD) Workshop. COST Action focused on developing clearer recommendations on air transport infrastructure and policy.

Strategic Policy Advisory Experience

- 2021–2022 Transportation policy advisor to the Israel Ministry of Transport, Member of the strategic national parking committee
- 2018–2022 Transportation policy advisor on parking policies to municipalities (Jerusalem, Tel Aviv and more) and government officials (Ministry of Transport, Ministry of Finance, Prime minister's office, Planning committees)

Industry Experience

- 2010–2016 The Jerusalem College of Engineering, System Analysis and Data Team Leader
Focusing on the integration of strategic data analysis to enhance operational and administrative functions. Led the design, deployment and continuous improvement of all information systems, significantly enhancing data-driven decision-making for middle and senior management. Collaborated extensively with internal stakeholders and external contractors to optimize system performance and align with organizational strategic goals

2007–2010 The Jerusalem College of Engineering, Information Systems and Data Analysis Engineer
Founded the Systems Analysis division within the IT department. Owned the process of gathering, extracting and compiling data across various sources using relevant tools such as SQL, Reporting Services and Tableau. Independently formatted, re-structured and validated data to ensure quality and readiness for analysis. Focused on performing detailed analytics to solve product and business problems, exploring and maintaining databases and ensuring data integrity

Professional Association Membership

The Institute for Operations Research and the Management Sciences (INFORMS)
Transportation Science and Logistics (TSL) Society at INFORMS
Operations Research Society of Israel (ORSIS)
The Israeli Association of Transportation Research

Computer Skills And Other Tools

Optimization Python (e.g., SciPy, Pyomo, Gurobi), Wolfram Mathematica.
Data Science R, Python (e.g., scikit-learn, tensorflow), SPSS, STATA.
Visualization GeoPandas, ArcGis, Qgis.
Database, BI SQL, SqlServer (including Reporting Services), ERD, Tableau.